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Integrated Discovery Case Study-1

Integrated Drug Discovery: Case Study 1

- Objectives:
  - **To identify and develop potent and selective agonist of PPARδ for treatment of muscular disease and kidney injury**

ELSEVIER
Bioorganic & Medicinal Chemistry Letters
Volume 27, Issue 23, 1 December 2017, Pages 5230-5234

Novel highly selective peroxisome proliferator-activated receptor δ (PPARδ) modulators with pharmacokinetic properties suitable for once-daily oral dosing

Bharat Lagu [a], Arthur F. Kluge [a], Ross A. Fredenburg [a], Effie Tozzo [a], Ramesh S. Senaiar [b], Mahaboobi Jaleel [b], Sunil K. Panigrahi [b], Nirbhay K. Tiwari [b], Narasimha R. Krishnamurthy [b], Taisuke Takahashi [c], Michael A. Patane [a]

ELSEVIER
Bioorganic & Medicinal Chemistry Letters
Volume 28, Issue 3, 1 February 2018, Pages 533-536

Highly selective peroxisome proliferator-activated receptor δ (PPARδ) modulator demonstrates improved safety profile compared to GW501516

Bharat Lagu [a], Arthur F. Kluge [a], Matthew M. Goddeeris [a], Effie Tozzo [a], Ross A. Fredenburg [a], Shekar Chellur [b], Ramesh S. Senaiar [b], Mahaboobi Jaleel [b], D. Ravi Krishna Babu [b], Nirbhay K. Tiwari [b], Taisuke Takahashi [c], Michael A. Patane [a]

ELSEVIER
Bioorganic & Medicinal Chemistry Letters
Volume 30, Issue 4, 15 February 2020, 126928

Addressing hERG activity while maintaining favorable potency, selectivity and pharmacokinetic properties of PPARδ modulators

Bharat Lagu [a], Ramesh S. Senaiar [b], Arthur F. Kluge [a], B. Mallesh [b], M. Ramakrishna [b], Raveendra Bhat [b], Michael A. Patane [a]

Selective PPARδ Modulators Improve Mitochondrial Function: Potential Treatment for Duchenne Muscular Dystrophy (DMD)

Bharat Lagu*, Arthur F. Kluge, Effie Tozzo, Ross Fredenburg, Eric L. Bell, Matthew M. Goddeeris, Peter Dwyer, Andrew Basinski, Ramesh S. Senaiar, Mahaboobi Jaleel, Nirbhay Kumar Tiwari, Sunil K. Panigrahi, Narasimha Rao Krishnamurthy, Taisuke Takahashi, and Michael A. Patane

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Publications:

Bioorg. Med. Chem. Lett. **2017**, *28*, 5230
Bioorg. Med. Chem. Lett. **2018**, *28*, 533
ACS Med. Chem. Lett. **2018**, *9*, 935
Bioorg. Med. Chem. Lett. **2020**, 30, 126928

Patents:

WO2016057660
WO2016057658
WO2017062468
WO2017180818

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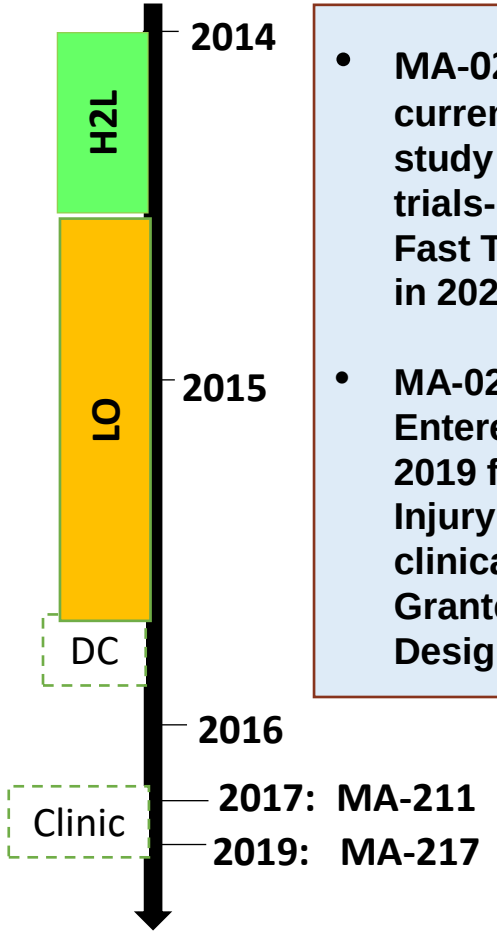
Identification of selective PPARδ agonist

- Objectives:
  - **To identify and develop potent and selective agonist of PPARδ for treatment of muscular disease and kidney injury**

- Project was divided into two stages
  - **Stage 1**: **Assay setup & lead identification**
  - **Stage 2**: **Lead optimization**

- Achievements
  - **Successfully developed two clinical candidate**
    - **DMD (primary indication)**
    - **Acute Kidney Injury (secondary indication)**



- **MA-0211 (ASP0367) is currently in Phase II / III study (MOUNTAINSIDE) trials- U.S. FDA Granted Fast Track Designation in 2020**
- **MA-0217 (ASP1128)- Entered clinical trial in 2019 for Acute Kidney Injury is in Phase-II clinical trials : U.S. FDA Granted Fast Track Designation in 2019**

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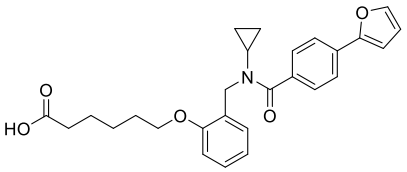
Hit to Lead compound

- **Hit compound 1 was cell potent and selective against isoforms (α and γ) . Low stability in MLM, High clearance and short T1/2 in mice → in vivo evaluation challenge. Limited SAR understanding.**

- **Polar groups/atoms in 'head' region was not tolerated. Non-polar substitution at α, β and γ positions improved clearance and exposures. Possibly, methyl substituents altered glucuronide formation (a major clearance mechanism) 'magic methyl' Compound-13 had desired balance potency and PK profile identified as lead molecule.**

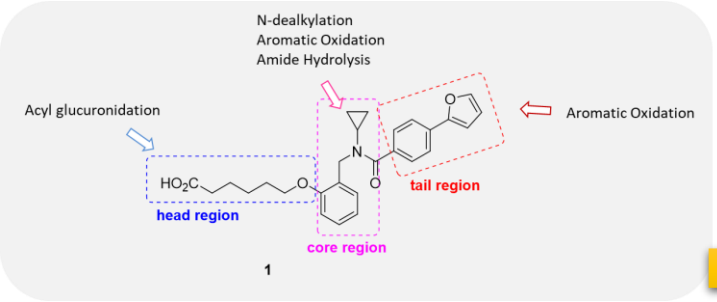
Hit

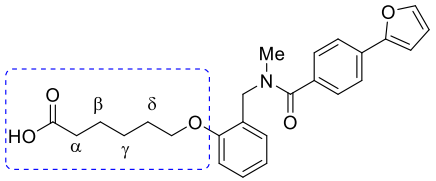


1
Hit Compound

MW / Clog P / tPSA		447 / 5.3 / 76.1
PPARδ EC_{50} (nM) Transactivation, CV1 cells		10
Thermo Sol pH 7.4 (μM)		252
Met. Stab. % remain @ 60 min	MLM	20
	HLM	65
IV PK (mice) 3 mpk	Beta T1/2 (h)	0.2
	C_0 (ng / mL)	1928
	Vss (L/kg)	1.6
	Cl (mL/min/kg)	195

~200% mouse hepatic blood flow (90 mL/min/kg)



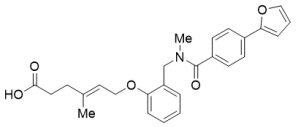


2
MW: 421 / Clog P: 4.7

PPARδ, EC_{50} = 13 nM
MLM, % rem@ 60 min: 56%
Cl (mL/min/kg) (i.v.3mpK), 44

Lead

Assay	Results
Human PPARδ	EC_{50} = 5 nM[a]; EC_{50} = 37 nM[b]
Human PPARα	EC_{50} >10,000 nM[a]; EC_{50} = 6100 nM[b]
Human PPARγ	EC_{50} >10,000 nM[a,b]
Mouse PPARδ	EC_{50} = 270 nM
Selectivity	• No activity in Eurofin PanLabs LeadProfilingScreen® (68 targets) upto 10μM. • No activity (up to 10mM) for androgen, progesterone or glucocorticoid receptors
Thermodynamic solubility	190 μM
Caco-2 permeability	A to B = 4.58E-05; B to A= 1.03E-04 (Efflux ratio 2.24)
CYP450 inhibition	>10 μM (CYPs 3A4, 2C9, 2C19, 2D6, 1A2)
hERG (patch clamp)	No inhibition upto 30 μM
Mutagenicity	Non-mutagenic in mini-Ames test
Mouse PK[d]	**1 mg/kg IV:** t½ = 5.8 h; **CL = 8.4 mL/min/kg**; Vss= 2.6 L/kg ; AUC = 1,980 ng*h/mL **10 mg/kg PO:** t½ = 4.3 h; Cmax = 11,100 ng/mL; AUC = 16,600 ng*h/mL; %F = 84
Rat Pk[e]	**3 mg/kg IV:** t½ = 4.5 h ; Vss= 0.9 L/kg ; **CL = 7.2 mL/min/kg**; AUC = 4,650 ng*h/mL **10 mg/kg PO:** t½ = 2.5h; Cmax = 3,805 ng/mL; AUC = 12,500 ng*h/mL; %F = 82
Monkey PK[f]	**3 mg/kg IV:** t½= 1.8 h ; **CL = 8 mL/min/kg**; Vss= 1.3 L/kg ; AUC = 6,200 ng*h/mL **10 mg/kg PO:** t½= 1.1 h: Cmax = 4,190 ng/mL; AUC = 6,050 ng*h/mL; %F = 29



13
MW: 433 / Clog P: 4.8

[A] Protein interaction assay; [b] Transactivation assay; [c] K_D for GW501516 in the SPR assay was 7.8±0.2 nM; [d] Male CD1 mice (n = 3); [e] Male Wistar rats (n = 3); [f] Male cynomolgus mokeys (n = 3)

Bioorg. Med. Chem. Lett. **2018**, *28*, 5230; *Proc. Natl. Acad. Sci.* **2017**, 114, E2563

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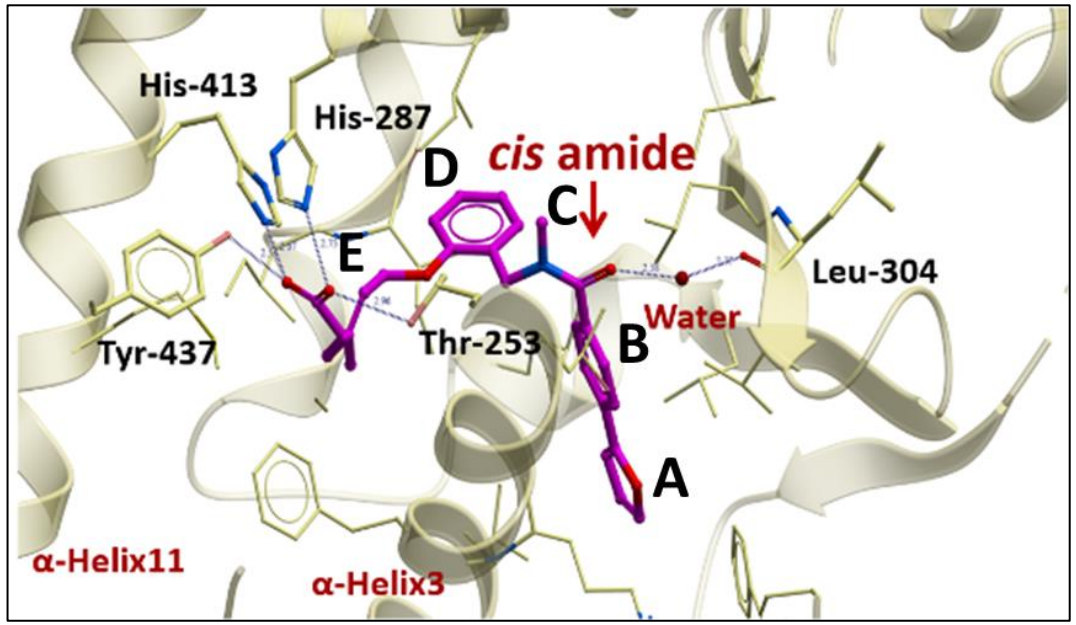
X-ray crystal structure

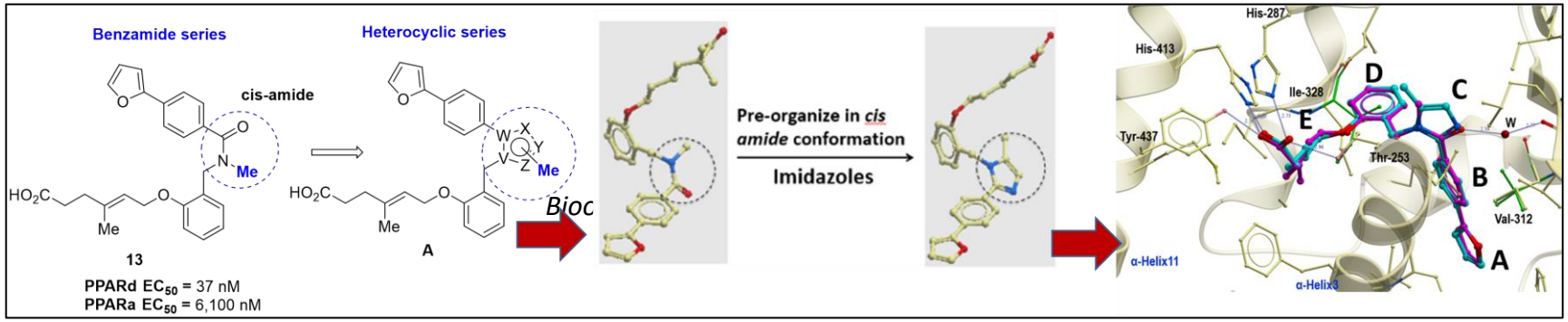
- Compound-13 bound to LBD of hPPRδ. Resolution: 2.0 Å
- The carboxylic end forms H-bond interactions with side chains of His287, His413, Thr253 and Tyr437.
- A lone water mediated interactions between the amide –CO group and Leu304 main chain atom.
- The ligand is surrounded by many hydrophobic residues like Leu294, Phe291, Leu303, Leu304, Val305, Val312.
- The hydrophobic contacts are strong at the entrance of the Y-shaped LBP near the furan binding.
- A closer analysis of X-ray structure revealed that amide bond was in *cis*-orientation.
- *trans*-amide are known to be thermodynamically more favoured, while *cis*-amide are uncommon (exception like proline).



Will a molecule in 'locked *cis*-amide' conformation have improved binding affinity for PPARδ ?

BIRTH OF HETEROCYCLIC SERIES

➢ *cis*-Amide ↔ 1,2-substituted heterocycle



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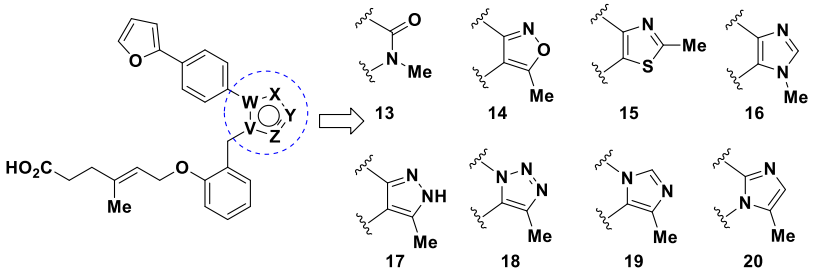
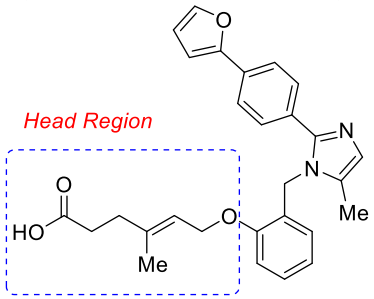
Heterocycles series optimization

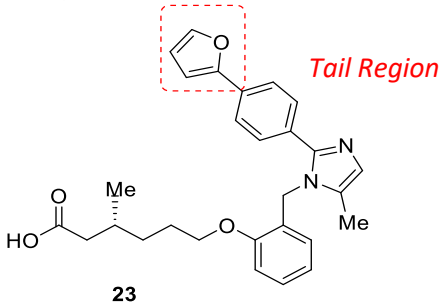
- **Potency improved (EC50 < 10 nM vs 13 ; except 19), All Compounds showed high CL (except -15), Compound-20 with good ADME properties selected for further optimization**

- **To address CL: Tail region' modification in Compound-21, Furan replacements no improvement, Substituting or replacing phenyl ring no improvement 'Head region' modification…… test 'magic methyl', Compound-23 with balance of potency and high selectivity was chosen for further optimization**

- **Compound-30 (MA-0204) had sub-nanomolar potency, very high isoform selectivity, reduced clearance and good plasma exposure. It was one of the contenders as a 'pre-clinical candidate'**

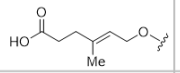
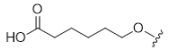
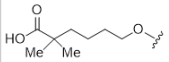
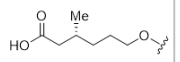






CompNo	PPARδ EC_{50} nM[a]	PPARα EC_{50} nM[a,b]	t½ (h)[c]	CL (mL/min/kg)[c]	AUC (ng*h/mL)[c]
13	37	6,100	2.3	15	3300
14	1.4	5,840	0.2	182	275
15	9.5	7,247	0.8	56	894
16	1.3	419	0.8	296	169
17	1.2	>10,000	0.4	230	380
18	8.4	>10,000	0.2	225	220
19	59	ND	0.7	719	66
20	1.0	1,230	0.5	164	305

[a]Transactivation [b]All the compounds, EC_{50} >10,000 nM for PPARγ; [c]Exposure data for compounds dosed i.v. at 3 mg/kg to CD-1 mice; ND = Not Determined

Comp No	Head region Modifications	PPARδ EC_{50} (nM)[a]	PPARα EC_{50} (nM)[a]	t½ (h)[b]	CL (mL/min/kg)[b]	AUC (ng*h/mL)[b]
20		1.0	1,230	0.5	164	305
21		0.7	6,965	0.3	270	180
22		4.5	ND	2.4	120	400
23		1.7	>10,000	1.4	73	650

Comp No	Top 'tail region'	PPARδ EC_{50} (nM)[a]	PPARα EC_{50} (nM)[a]	t½ (h)[b]	CL (mL/min/kg)[b]	AUC (ng*h/mL)[b]
23	2-furyl	1.7	>100,000	1.4	73	650
24	4-Cl-3-F	0.8	69,000	4.3[c]	38[c]	440[c]
26	4-Cl	5.5	>100,000	1.4[c]	22[c]	752[c]
27	4-Me	20	78,500	ND	ND	ND
28	4-$OCHF_2$	13.5	15,060	ND	ND	ND
29	4-F	39	>100,000	ND	ND	ND
30 (MA-0204)	4-OCF_3	0.4	6,900	3.3	25	1250

[a]Transactivation assay[15] [b]Exposure data for compounds dosed i.v. at 3 mg/kg to CD-1 mice; [c]Exposure data for compounds dosed i.v. at 1 mg/kg to CD-1 mice; ND = Not Determined. All the compounds showed EC_{50} >10,000 nM for PPARγ.

ACS Med Chem **2018**, *9*, 935

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MA-0211 improves mitochondrial function modulate target genes and is efficacious in MDX mice model

In DMD patient cells and mouse muscle cells, MA-0211

– Increases expression of genes and proteins that enhance mitochondrial function

– Increases numbers of mitochondria in muscle cells (biogenesis)

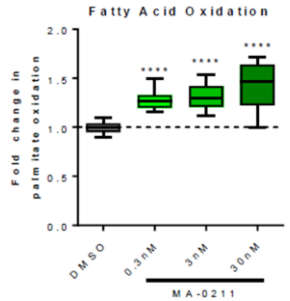
MA-0211 increases PPARδ target gene expression: Potent engagement of PPARδ target genes in DMD myotubes demonstrated with MA-0211, after treatment for 24h.

Efficacy: In mdx mouse model of DMD, administration (orally) of MA-0211 orally for 5-7.5 weeks,

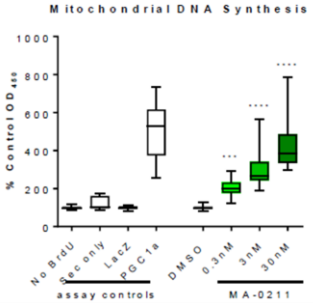
– Significant increase in the endurance of mice while running on a treadmill.

– Reduced muscle inflammation and degeneration, and decreased diaphragm scarring.

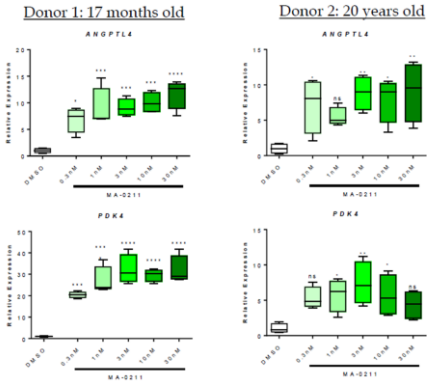
-Reduced creatine kinase levels in mice serum and improved their cardiac and respiratory function relative to untreated mdx mice

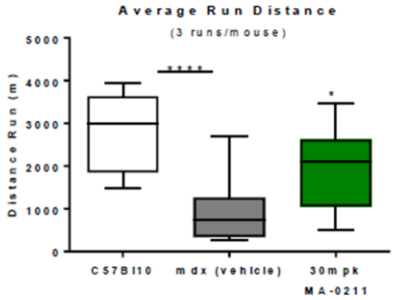


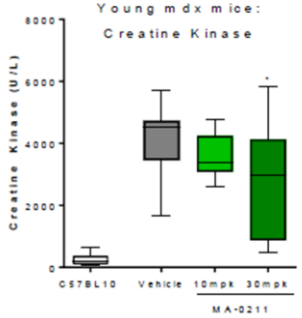
Fatty acid oxidation was increased after 24 hours of treatment in DMD myotubes from 17 month old donor (measured using Seahorse XF analyser).

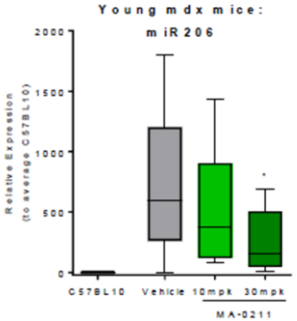


Mitochondrial DNA synthesis measured by BrdU incorporation, a measure related to mitochondrial biogenesis, increased with 72 hour treatment









Parent Project Muscular Dystrophy (PPMD) 2017 Connect Conference Poster by Mitobridge (Www.mitobridge.com)

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MA-0211 and MA-217: Clinical candidate

- **MA-0211** (MTB-1), a highly selective, oral PPARd agonist reproducibly demonstrated efficacy in multiple studies with DMD animal models via improvement in mitochondrial function and metabolism.
- Studies in DMD patient muscle cells and mdx mice demonstrate the compound is active at dose levels that are well-tolerated.
- Mitochondrial/metabolic benefits are independent of dystrophin mutation and offer potential for combination with other treatments.
- **MA-0211 (ASP0367)** is currently in Phase II / III study (MOUNTAINSIDE) trials- U.S. FDA Granted Fast Track Designation in 2020
- **MA-0217 (ASP1128)**-Entered clinical trial in 2019 for Acute Kidney Injury is in Phase-II clinical trials : U.S. FDA Granted Fast Track Designation in 2019

A Study to Assess the Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of MA-0211 in Healthy Adult Subjects Including a Food Effect Cohort

https://clinicaltrials.gov/

ClinicalTrials.gov Identifier: NCT03682484

Study to Evaluate the Efficacy of ASP1128 (MA-0217) in Subjects at Risk for Acute Kidney Injury Following Coronary Artery Bypass Graft (CABG) and/or Valve Surgery

ClinicalTrials.gov Identifier: NCT03941483

Publications:

***Bioorg. Med. Chem. Lett*. 2017, *28*, 5230**
***Bioorg. Med. Chem. Lett*. 2018, *28*, 533**
***ACS Med. Chem. Lett.* 2018, *9*, 935**
***Bioorg. Med. Chem. Lett*. 2020, 30, 126928**

Patents:
WO2016057660
WO2016057658
WO2017062468
WO2017180818.

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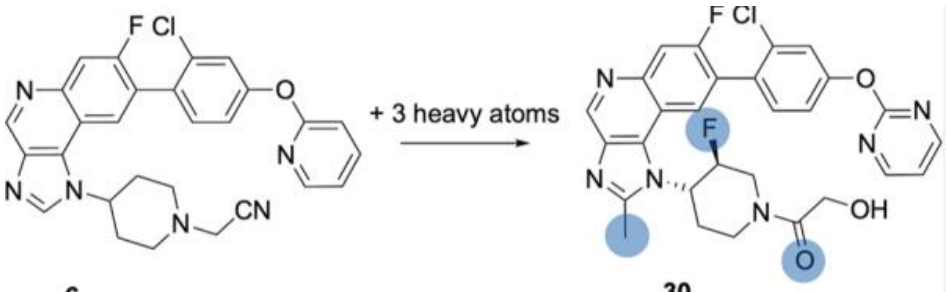
Integrated Discovery Case Study-2

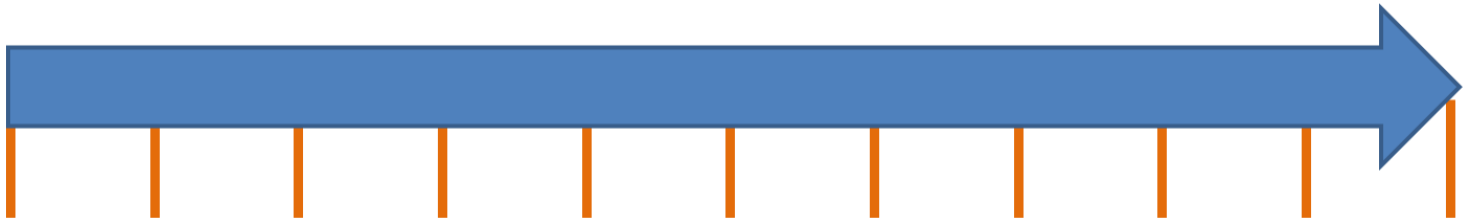
Integrated Drug Discovery: Case Study 1- MAP855

- Objective:
  - **To develop a first-in-class, selective ATP competitive MEK enzyme inhibitor for the treatment of cancer refractory to current treatment**

Differentiation of MEK-ATPi vs. Allosteric-MEKi:

- Preventing MEK induction
- Sustained PD modulation
- Equipotency on MEK-wt and MEK-mutants
- Exploring opportunities towards RAS-mutant cancer





Oct 2009
Start

Lead

Mar 2014
Candidate

- Duration : 5 yrs.
- 6 series pursued (3 patents)
- 1 Publication

Ref: J. Med. Chem. 2022, 65, 5, 4350–4366

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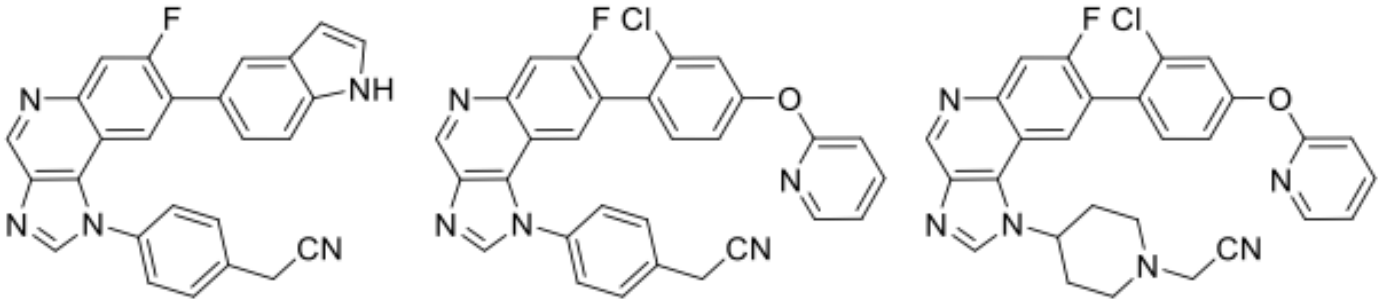
Hit identification: Compound 4 to Tool Compound 6

- HTS campaign using MEK1-ERK2 cascade assay resulted in moderately potent hit (compound 4) with a promising kinase selectivity profile, no activity on BRAF or ERK2 and a hint of cellular activity.

Table 1. Enzymatic and Cellular Potencies of 4–6



4 5 6

	MEK1 IC_{50} [μM]	A375 pERK EC_{50} [μM]	kinase selectivity[a]
4	2.6	5.8	0/7 of 37
5	0.1	0.3	0/3 of 37
6	0.07	0.4	0/2 of 64

[a]Off-target kinases with IC_{50} below 1 and 5 μM in the Novartis in-house kinase panel.

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Hit 4 to Tool Compound 6

- Optimization of back pocket, fluorine substitution and nitrile lead to compound 5. The first example with the desired activity profile, showing good kinase and pathway selectivity and no shift between A375 V211D versus WT pERK and proliferation IC50

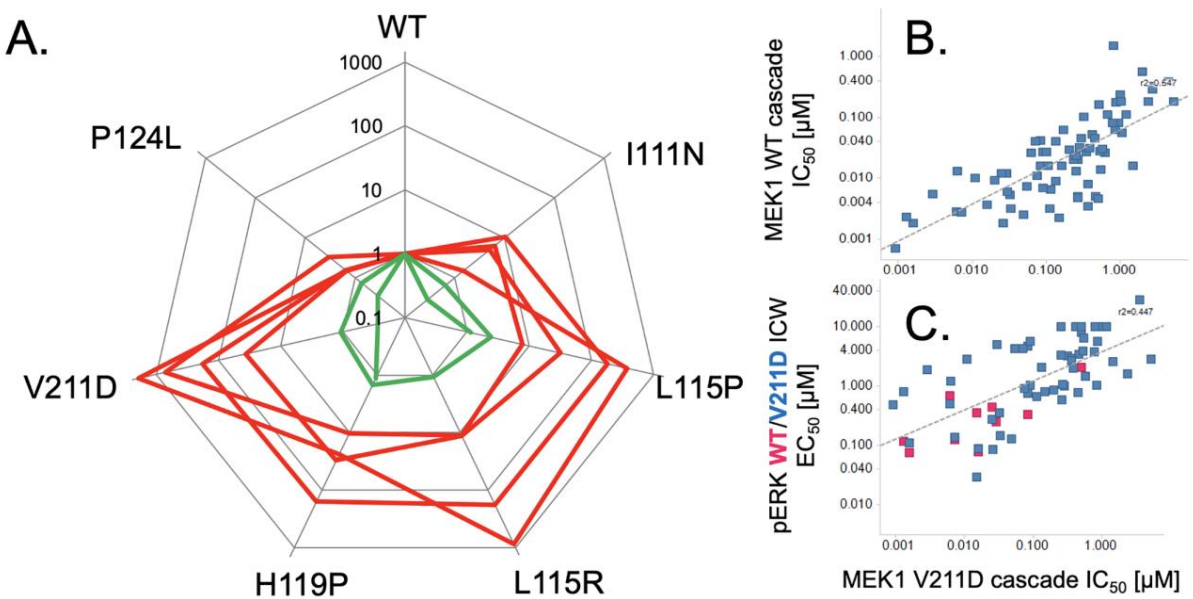
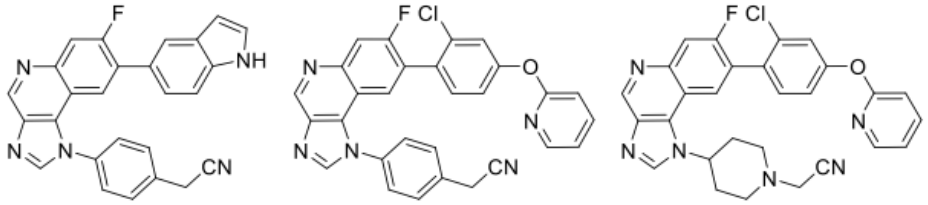


Table 1. Enzymatic and Cellular Potencies of 4–6



4 5 6

	MEK1 IC_{50} [μM]	A375 pERK EC_{50} [μM]	kinase selectivity[a]
4	2.6	5.8	0/7 of 37
5	0.1	0.3	0/3 of 37
6	0.07	0.4	0/2 of 64

[a]Off-target kinases with IC_{50} below 1 and 5 μM in the Novartis in-house kinase panel.

- **A) Resistance mutant profile of representative ATP-competitive (green) versus clinical allosteric MEK1/2 inhibitors (red). The radar plot shows the ratio of mutant versus WT EC50 in the A375 + MEK1 proliferation assay for several resistance alleles.**
- **(B) Correlation plot of the biochemical MEK1 WT versus MEK1 V211D cascade assay IC50.**
- **(C) Correlation plot of the biochemical MEK1 V211D IC50 versus cellular A375 pERK EC50 (WT in red, V211D in blue).**

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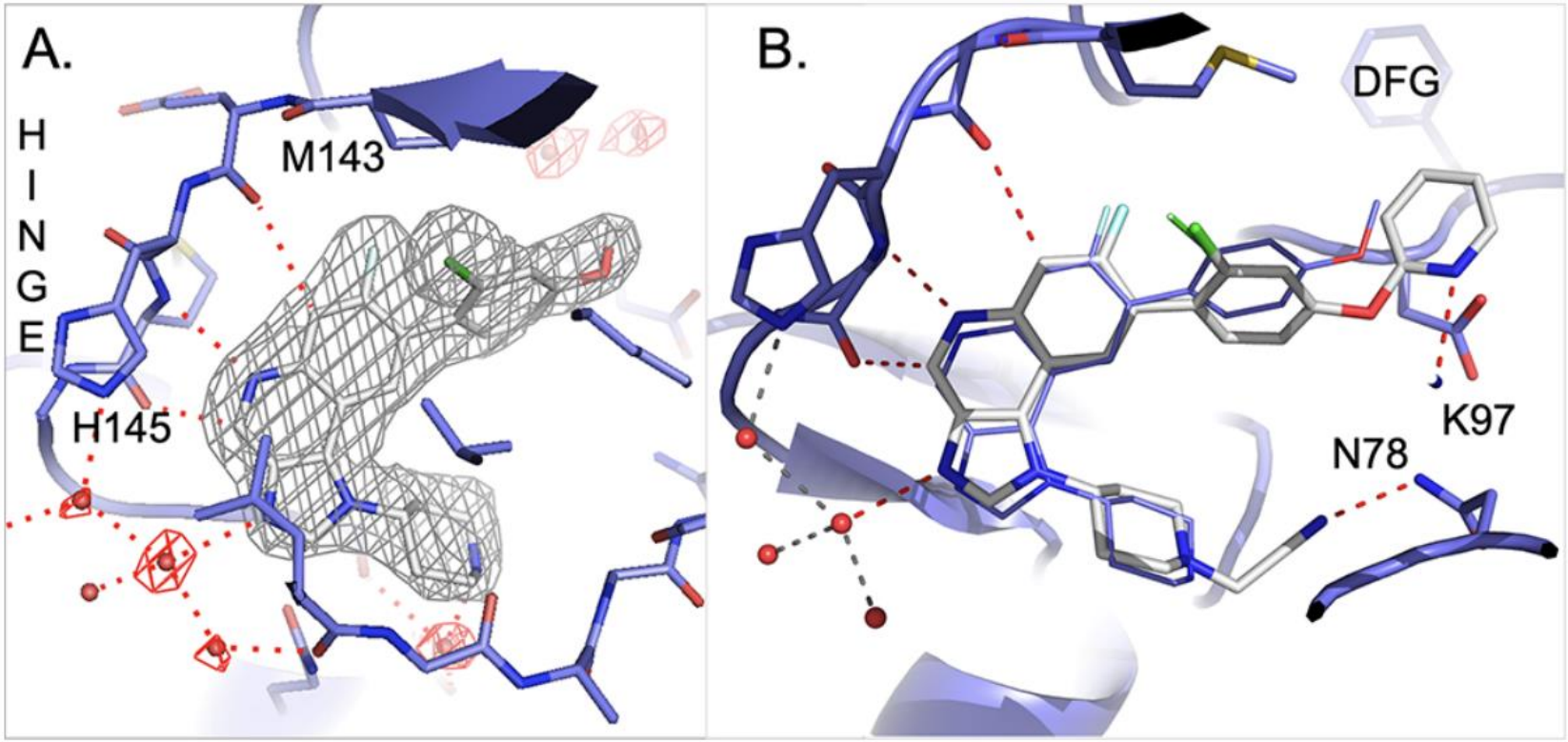
X-Ray and Optimization Strategy

- Analysis of the cocrystal structure of 7 with MEK1 (Figure 4A) revealed several details that helped rationalize the observed SAR.



- (A) Crystal structure of 7 bound to MEK1 at 2.1 Å resolution. Ligand and critical residues of MEK1 are shown as sticks the 2Fo−Fc electron density map of the ligand, as well as close-by waters are contoured at 1.5σ. (B) Modeled binding mode of 6 in MEK1.7 (PDB 7PQV) shown as sticks on top of the X-ray ligand (lines). Hydrogen bonds are shown as red dotted lines

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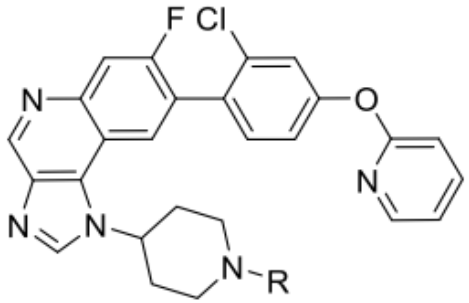
Solubility Optimization

- Stacking and shape complementarity of the core contribute to the high crystal lattice energy. Therefore, systematically varied the 2- and the 8-aryl ortho substituent to modulate solubility with a set of molecular matched pairs (MMP) 10–22

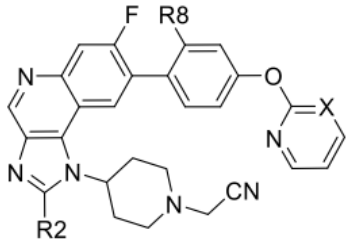
Table 2. ADME Properties of Compounds 6, 8, and 9



	R	pK_a	solubility [μM][b]	log P	PAMPA[c]
6	CH_2CN	4.1	ND[a]	5.3	−4.2
8	2-EtOH	6.9	67	4.6	−4.4
9	H	8.4	47	4.8	−4.5

[a]Could not be measured due to precipitation. [b]HT solubility at pH 6.8. [c]logPAMPA.

Table 3. ADME Properties of Compounds 6, 10–22



	R2	X	R8	solubility[a]	log P	PAMPA[c]
6	H	C	Cl	ND[b]	5.3	−4.2
10	Me	C	Cl	<4/175	5.1	−4.1
11	MeOH	C	Cl	<4/<4	4.9	−4.5
12	1-EtOH	C	Cl	<4/22	6.8	−4.7
13	H	C	OMe	ND[b]	5.3	−4.2
14	Me	C	OMe	<4/22	4.6	−3.9
15	MeOH	C	OMe	<4/7	4.4	−5.0
16	1-EtOH	C	OMe	<4/9	4.4	−4.6
17	H	N	Cl	ND[b]	3.0	−4.5
18	Me	N	Cl	11/106	3.3	−4.7
19	MeOH	N	Cl	12/6	3.5	−5.3
20	1-EtOH	N	Cl	<4/22	6.8	−4.7
21	Me	N	OMe	33/221	2.3	−4.5
22	1-EtOH	N	OMe	31/42	3.2	−6.9

[a]HT solubility at pH 6.8/FaSSIF in μM. [b]Could not be measured due to precipitation. [c]Log PAMPA.

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Cyp3A4 and HLM Optimization

- Another rationale for the introduction of a 2-Me (18) or junction-Me (24) group was to stabilize the proposed bioactive conformation of 6, and indeed both modifications consistently improved activity.

- Unfortunately, we also observed that stabilizing the closed conformation had a detrimental effect on the Cyp3A4 activity, for example, in the matched pairs around 6 (X = N), 18, and 24, the Cyp3A4 IC_{50} decreased from 13 to 1.5 to 0.6 µM, with a concomitant decrease in liver microsome (LM) stability.

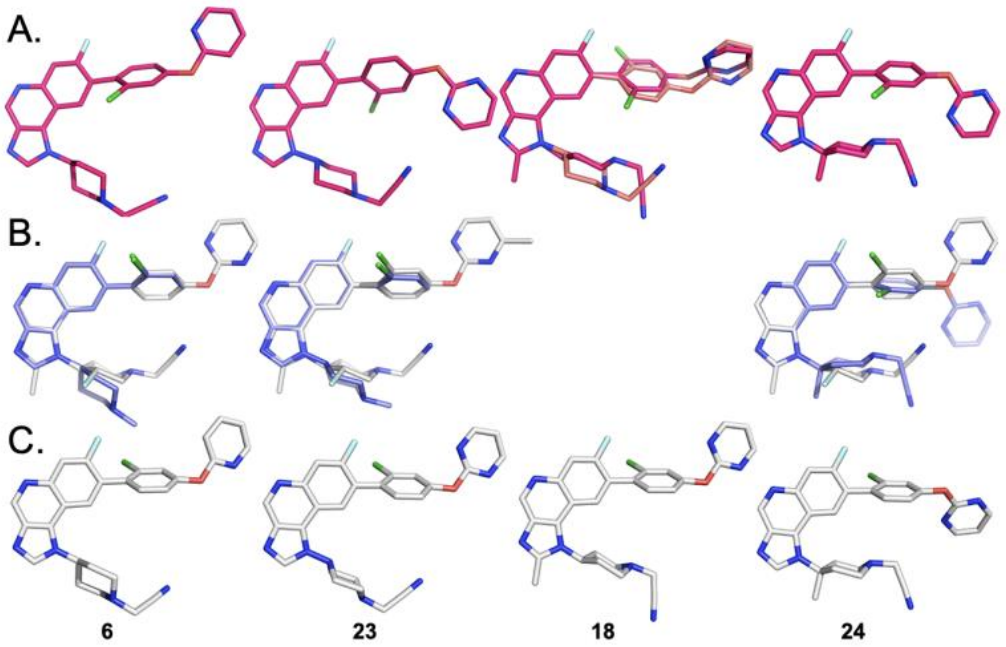
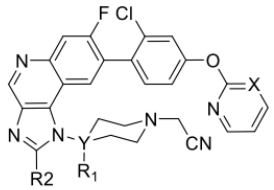


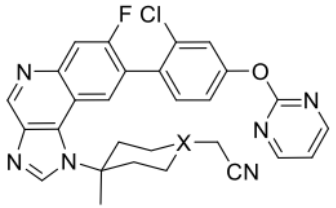
Table 4. Impact of R1 and R2 on Conformation



	R1	R2	Y	X	NMR open-closed[a]	QM open-closed[b]	A375 pERK IC_{50} [µM]	log P
6	H	H	C	C	>99:1	97:3	0.4	3.0[c]
23		H	N	N	>99:1	99:1	0.12	3.2
18	H	Me	C	N	~25:75	63:37	0.06	3.3
24	Me	H	C	N	~5:95	8:92	0.08	4.2

[a]Distribution of the open/closed conformation of piperidine/piperazine by solution NMR. [b]QM calculations. [c]Log P of 8′-pyrimidine MMP 17.

Table 5. Impact of Piperidine N on Clearance



	X	log P	Cyp3A4 IC_{50} [µM]	m/r/h LM	mouse ivCL
24	N	4.2	1.5	H/H/H	35
25	C	4.6	<0.5	M/ND/H	6

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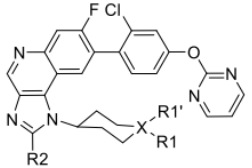
Impact of Exposed Polarity on Cyp3A4 Activity

- **MetID analysis revealed oxidation of the piperidine side chain to be a major clearance route**
- **The first hint that exposed polarity in this region could tune out Cyp3A4 came from a set of carbinamines 26–28**

Table 6. Impact of Exposed Polarity on Cyp3A4 Activity



	R2	X	R1	R1′	log *P*	Cyp3A4 IC_{50}[a]	m/r/h LM[b]
18	Me	N	CH_2CN		3.3	1.5	M/M/H
26	Me	C	CH_2CN	NH_2	4.2	1.2	M/M/H
27	Me	C	NH_2	CH_2CN	2.6	12.1	L/L/M
28	H	C	NH_2	CH_2CN	3.3	5.7	L/L/M

[a]In μM. [b]Upper (H), middle (M), and lower (L) 33 percentiles of the Novartis LM clearance data.

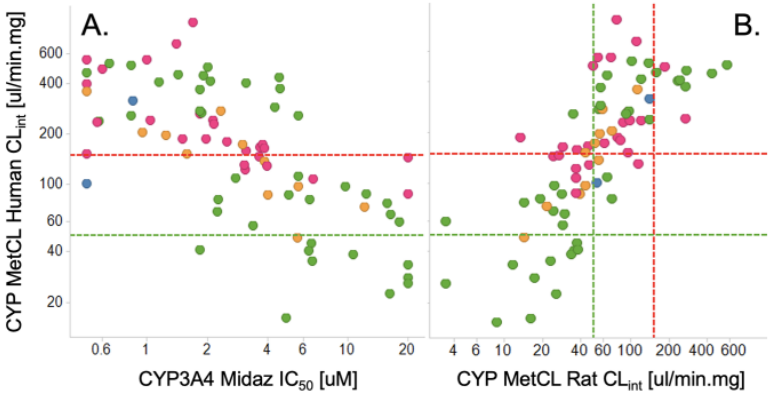


Figure 6. Correlation of human with rat microsome liver clearance (B) and Cyp3A4 inhibition (A). R1 subseries are color coded (red: piperidine nitriles, orange: cyclohexyl acetonitriles, blue: piperazines, and green: non-nitriles), the lines show the upper (red), and lower (green) 33 percentiles of the Novartis LM clearance data.

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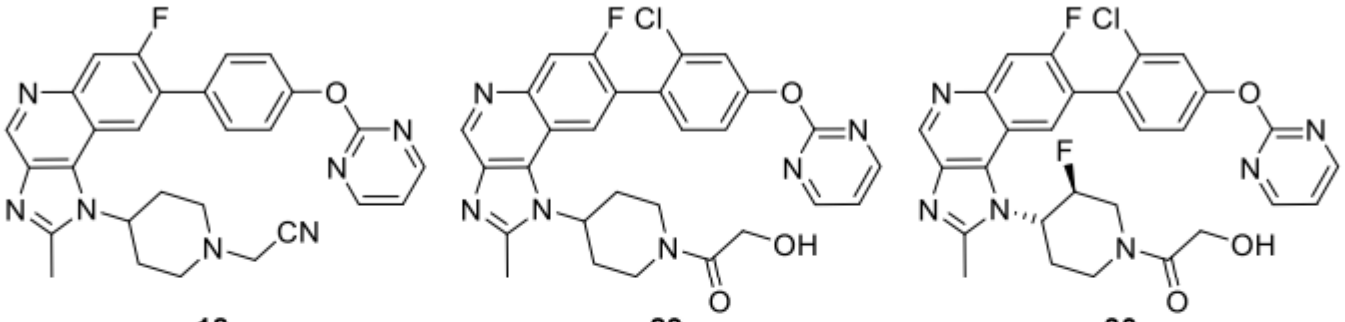
Cyp3A4 and HLM Optimization

- **Nitrile Replacements: replacing the nitrile acceptor (e.g., sulfonamides, ureas/amides, small heterocycles). From this exercise, hydroxy-acetamide 29 emerged with the best overall profile combining potency and a favorable physicochemical and Cyp3A4/HLM profile.**

- **Compound 30 provide overall balance in property and potency**

Table 7. Nitrile Replacement SAR



18 **29** **30**

	pERK EC50 [nM]	log *P*	PAMPA[a]	Cyp3A4 IC_{50} [uM]	m/r/h LM[b]	rat PK	
						rat iv CL[ml/min.kg]	% *F*
5	250	6.8	H	<0.5	H/M/H	8	5
18	60	3.3	H	1.6	M/M/H	16	0
29	50	2	L	11	L/L/L	34	37
30	5	1.6	M	7.7	M/L/M	35	65

[a]logPAMPA. [b]Upper (H), middle (M) and lower (L) 33 percentiles of the Novartis mouse - rat - human LM clearance data logPAMPA.

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Binding mode of 30: Modelling Insight

- **Compound 30 is characterized by a favorable physicochemical profile with medium to good permeability, decent solubility from crystalline material and low log P/D. Its moderate clearance. In rodents, 30 showed good oral bioavailability and medium clearance Cyp3A4 inhibition of 8 µM translated to medium to low LM**

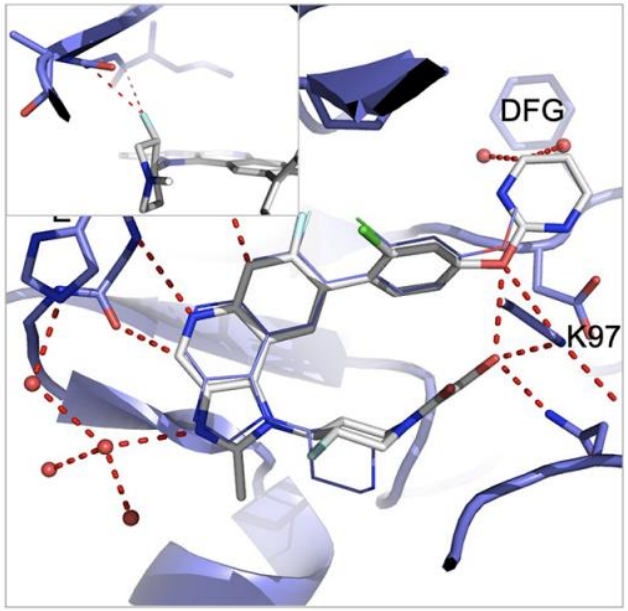

Figure 7. Modeled binding mode of **30** in the MEK1.7 co-xtal structure (PDB 7PQV) shown as sticks on top of the X-ray ligand (lines). Hydrogen bonds are shown as red dotted lines. The inset shows the orthogonal carbonyl fluorine interactions of the 3-F-piperidine with the P-loop Leu74 and Gly75 carbonyls.

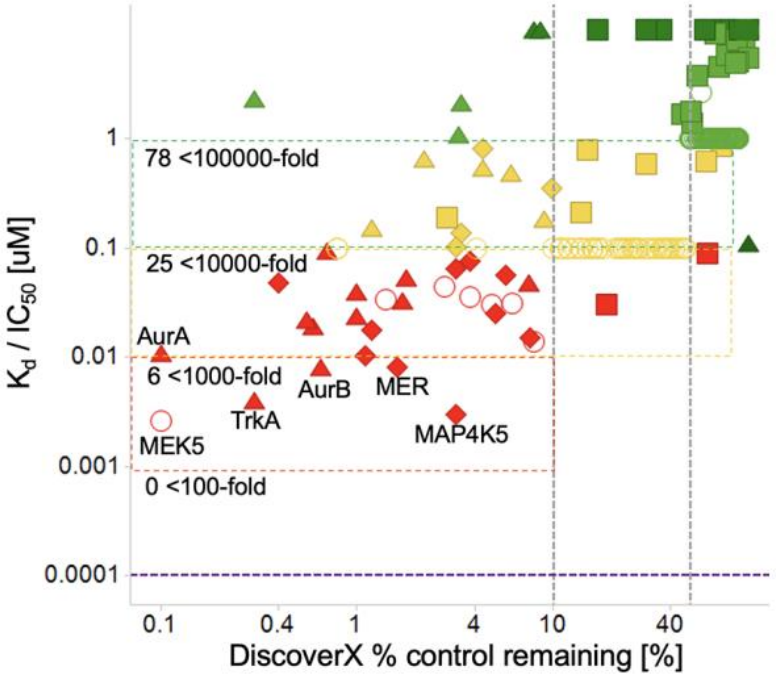

Figure 8. Biochemical selectivity of **30**. All targets with <10% ctrl in the DiscoverX panel were followed up with biochemical IC_{50} whenever available or K_d (square: inhouse IC_{50}, diamond: Invitrogen IC_{50}, triangle: Millipore IC_{50}, circle: DiscoverX K_d). IC_{50}/K_d are plotted *versus* the DiscoverX remaining control @ 1 μM **30**. Off-targets with 10–50% were set to 0.1 μM, off-targets with >50% were set to 1 μM IC_{50}. Targets below 10 nM K_d/IC_{50} are labeled.

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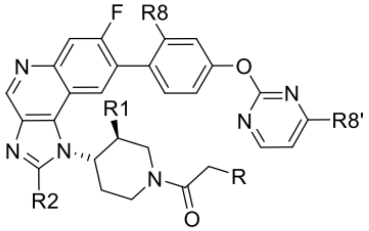
Compound 30: MAP855

- **Compound 30 have moderate clearance. In rodents, 30 showed good oral bioavailability and medium clearance Cyp3A4 inhibition of 8 µM translated to medium to low LM clearance**

Table 8. Deconstructive Single Atom Changes around 30



	R	R1	R2	R8	R8′	pERK[a]	PAMPA[b]	log *P*	Cyp3A4[c]
30	OH	F	Me	Cl	H	5	M	1.6	8
29	OH	H	Me	Cl	H	50	L	2	11
31	OH	F	H	Cl	H	340	L	2.3	>20
32	H	F	Me	Cl	H	64	ND	ND	ND
33	OMe	F	Me	Cl	H	109	H	3.2	5
34	OH	F	Me	Cl	Me	7	M	2.6	2
35	OH	F	Me	Me	H	19	M	2	>20

[a]A375 pERK EC50 [nM]. [b]logPAMPA. [c]Cyp3A4 IC50 [uM].

Table 9. Pharmacokinetic Profile of 30 in Mouse, Rat, and Dog

	mouse	rat	dog
CL [mL/min·kg]	32	35	22
V_{ss} [l/kg]	2.6	2.0	1.8
AUC po d.n. [µM·h][a]	0.4	0.6	1.4
Oral BAV [% *F*][a]	44	65	100

[a]Dosed as solution formulation.

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MAP855: PK/PD and efficacy

- In a PK/PD experiment in A375 tumor-bearing mice, 30 dosed at 30 mg/kg showed a robust inhibition of pMEK, pERK, DUSP6, and SPRY4 up to the 8 h time point whose free exposure corresponded to the pERK EC90 (Figure 9A,B).
- In a A375 tumor-bearing mice efficacy study, 30 dosed at 30 mg/kg b.i.d. achieved comparable efficacy to trametinib dosed at the mouse MTD without any body weight loss (Figure 9C). In line with the PK and our efficacy model (time over EC90 as driver), 30 mg/kg b.i.d. was superior to 60 mg/kg qd.

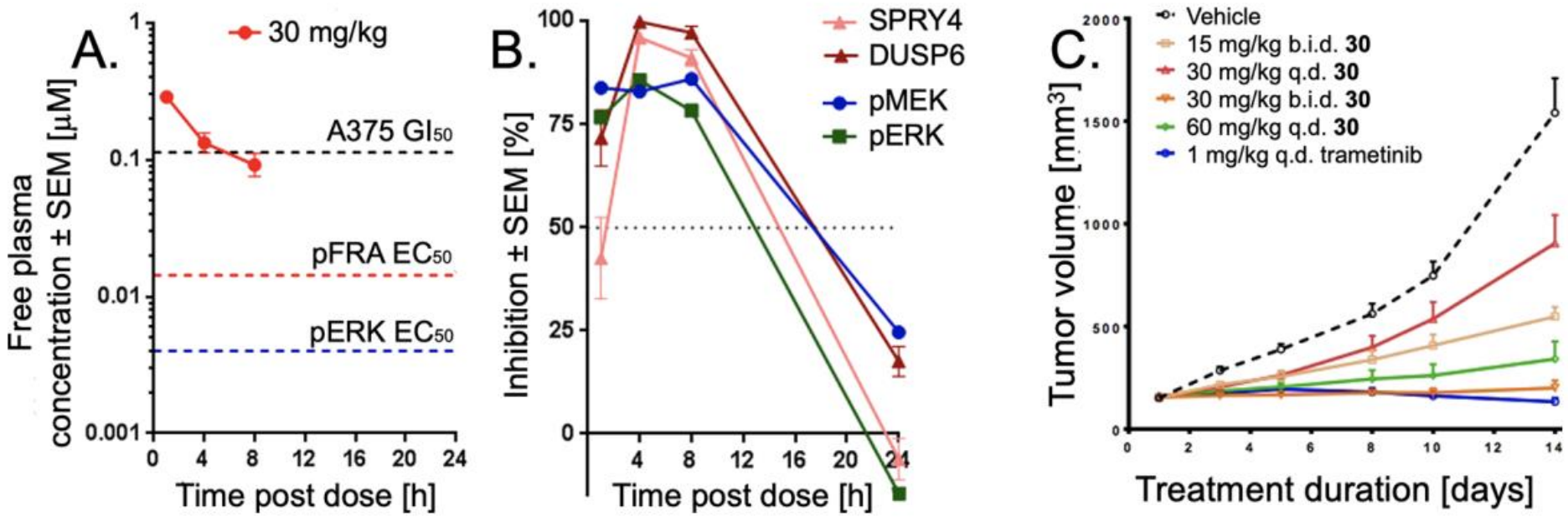

Figure 9. A,B) PK–PD of **30** in A375 tumor-bearing mice dosed at 30 mg/kg. (C) Efficacy of **30** in A375 tumor-bearing mice *versus* trametinib as the control. % T/C and % TGI values are given in Table S9.

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Key success

- Identify Development Candidates within the budget and timeline
- Develop robust synthetic process for scaling up the compounds
- Identify IND candidates with clear Intellectual Property right
- Be mindful of cost of goods, BCS class of NCE and Predicted Human PK